

```
fig, ax = plt.subplots()
❶ ax.plot(dates, highs, color='red', alpha=0.5)
  ax.plot(dates, lows, color='blue', alpha=0.5)
❷ ax.fill_between(dates, highs, lows, facecolor='blue', alpha=0.1)
--snip--
```

The `alpha` argument controls a color's transparency ❶. An `alpha` value of 0 is completely transparent, and a value of 1 (the default) is completely opaque. By setting `alpha` to 0.5, we make the red and blue plot lines appear lighter.

We pass `fill_between()` the list `dates` for the x -values and then the two y -value series `highs` and `lows` ❷. The `facecolor` argument determines the color of the shaded region; we give it a low `alpha` value of 0.1 so the filled region connects the two data series without distracting from the information they represent. Figure 16-5 shows the plot with the shaded region between the `highs` and `lows`.

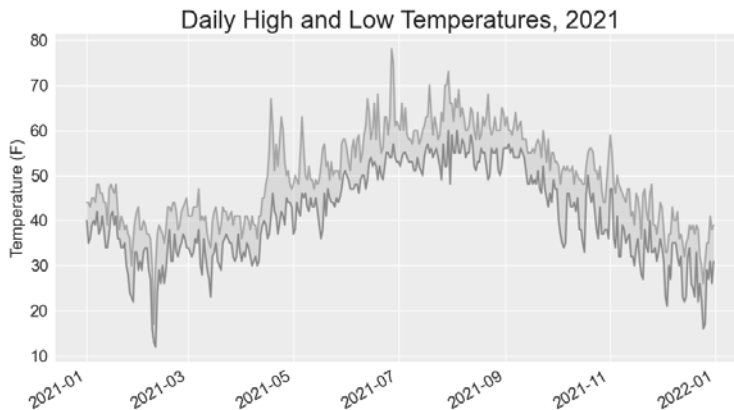


Figure 16-5: The region between the two datasets is shaded.

The shading helps make the range between the two datasets immediately apparent.

Error Checking

We should be able to run the `sitka_highs_lows.py` code using data for any location. But some weather stations collect different data than others, and some occasionally malfunction and fail to collect some of the data they're supposed to. Missing data can result in exceptions that crash our programs, unless we handle them properly.

For example, let's see what happens when we attempt to generate a temperature plot for Death Valley, California. Copy the file `death_valley_2021_simple.csv` to the folder where you're storing the data for this chapter's programs.